3 · *What Drives Species Distributions?*

The most obvious questions in biogeography, and the starting points for this book, are: what are we observing (i.e. which species, communities, or ecosystems?)? Where in space and time? Why are organisms distributed where they are? The quest to answer these questions is an age-old one (Guisan and Thuiller, 2005; Franklin, 2010a), but which really took off scientifically with the eighteenth and nineteenth centuries' first biogeographers (i.e. Von Humbolt, De Candolle, Darwin, and Wallace). Since the steady rise of ecological research during the twentieth century, explaining and understanding the distribution of biodiversity at various spatial and temporal scales has continued to be an important field of macroecological and biodiversity research (Lomolino et al., 2010; McGill, 2010). Numerous new or refined theories were proposed (metapopulation dynamics, e.g. Hanski and Gilpin, 1997; neutral theory, Hubbell, 2001; metabolic theory, Brown et al., 2004) in which geographic space was explicitly considered (Moloney and Jeltsch, 2008). This ultimately fostered the development of predictive models of species and biodiversity (Côté and Reynolds, 2002; Guisan et al., 2013).

This chapter does not attempt to review every single step in this long history, nor does it aim to provide an exhaustive review of all the theoretical aspects of species' niches and distributions. Instead, we aim to present the theories and findings most relevant to habitat suitability modeling. However, and although habitat suitability remains the major principle behind this type of model, we will begin with a more general biogeographical perspective. First, we will present the three specific key drivers of species distributions (3.1), then introduce each of them more detail in the following sections: speciation, dispersal, species pools and neutral theory (3.2), the abiotic environment: habitat and fundamental niche (3.3), and the biotic environment: species interactions, community assembly and the realized niche (3.4). Habitat and niche issues will then be further discussed in more detail in Chapters 4 and 5, from the angle of habitat suitability modeling. The aim here is to introduce the basic

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knowledge around these concepts that is useful for habitat suitability modeling rather than reviewing them exhaustively.

3.1 The Overall Context: Dispersal, Habitat, and Biotic Filtering

When building models to explain and predict the distribution of organisms we necessarily need to ask the same questions as the early biogeographers. It is now clear that **three main conditions** need to be met for a species to occupy a site and maintain populations (Figure 3.1, see Pulliam, 2000; Lortie et al., 2004; Soberón, 2007):

(i) the species has to reach the site, i.e. to access the region (Barve et al., 2011) and disperse there;
(ii) the abiotic environmental conditions must be ecophysiologically suitable for the species;
(iii) the biotic environment (interactions) must be suitable for the species.

The first condition is a matter of species **dispersal** capacity from those areas previously occupied by the species. It includes the biogeographic history of the species, and thus all factors limiting its distribution from the place where it first originated, such as barriers to migration, biotic and abiotic dispersal vectors, etc. (Section 3.2).

The second condition is the matter of abiotic **habitat suitability** for the target species, which implies that the combination of abiotic environmental variables at the site – often referred to as environmental suitability – is included in the environmental conditions that a species needs to grow and maintain viable populations, i.e. its environmental niche (sensu Hutchinson, 1957) and constitutes the basis of the habitat suitability modeling approach (Guisan and Zimmermann, 2000) presented in this book (Section 3.3 and Chapter 4).

The third condition concerns **biotic interactions**, i.e. interactions with other organisms, either positive (commensalism, mutualism) or negative (competition, predation), which themselves are dictated by the environment through their influence on all organisms in the local community (Section 4.4). This component can also include top-down environmental constraints on communities, if applicable, such as the idea that whole communities and ecosystems may also have their species composition limited by some form of environmental carrying capacity (Del Monte-Luna et al., 2004; but see Buckley et al., 2010; see Guisan and Rahbek, 2011). This may also involve some species engineering the

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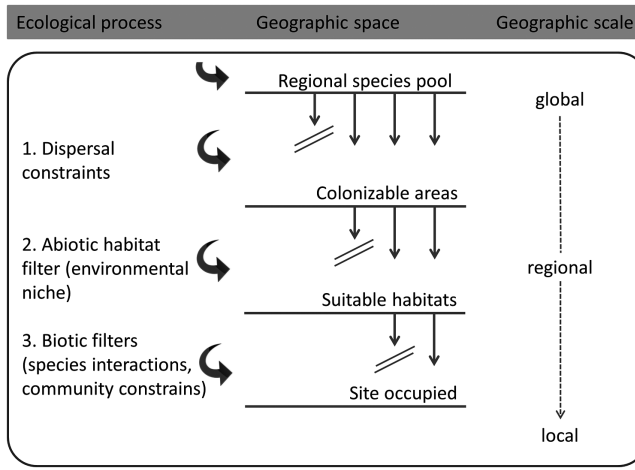


Figure 3.1 Hierarchical view of the three main influences determining species occurrence at a given site: dispersal limitation, abiotic habitat filtering, and biotic filters, with corresponding geographic space at the successive scales, from global to regional to local; see also Pulliam (2000), Lortie et al. (2004), and Soberón (2007). Note that these processes may not act in the sequence and hierarchy suggested, and stochastic or neural processes might counteract some of these deterministic processes and add randomness to the assemblages (modified from Guisan and Rahbek, 2011, with permission). See Figure 3.2. for a more balanced view of the three drivers operating simultaneously.

environment for other species, such as trees providing favorable canopy cover conditions for shade-tolerant species (Nieto-Lugilde et al., 2015).

As we will see later, assessing habitat suitability constitutes the core of the whole HSM approach that is the focus of this book (Guisan and Zimmermann, 2000). However, it is also intimately related to, and depends on, the other two drivers: dispersal limitations and biotic interactions (i.e. through environmental engineering), which can be added as additional steps to constrain habitat suitability predictions. What is less clear are the relative roles these three drivers play in shaping species distributions and assemblages. Addressing this issue primarily requires gathering knowledge about these processes along geographic and environmental gradients. This, however, cannot be achieved with the sole use of *in situ* observations; it requires complementary approaches such as *in situ* measurements (e.g. ecophysiological) along key environmental gradients, reciprocal transplant experiments, removal experiments in natural or artificial communities, common garden experiments, or controlled *ex situ* experiments in the laboratory (e.g. ecotons). For instance, one can

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move high-elevation plants to low elevations to test the hypothesis that, despite the fact that they can grow and reproduce well at low elevations, they can systematically be excluded by more competitive plants, as well as determining to what extent phenotypic differences between local populations along such gradients have a genetic basis (see Clausen et al., 1948; Hautier et al., 2009). The data and results collected using these approaches constitute the basis of the knowledge used to untangle the three main drivers of species distributions and can be used to build ecologically more meaningful HSMs (Austin, 2002).

This triple influence – dispersal, niche, and biotic – shaping species distributions, and their interactions, can be viewed schematically as separate ensembles defined by specific boundary conditions (Soberón, 2007). Suitable conditions for a species according to all three factors are found at the intersection of the three ensembles (Figure 3.2, case 1). We discuss these three factors and their interactions in more detail in the next three sections, including their role in explaining local species occurrences and their implications for modeling species distributions.

Individual populations of a species may be present in suboptimal situations (cases 2–4 in Figure 3.2) outside the intersection of suitable conditions (i.e. case 1 in Figure 3.2), either in suitable environments in which the species would usually be excluded by biotic interactions (Figure 3.2, case 2), or in non-suitable sink environments colonized through high propagule pressure (source–sink dynamics, see Pulliam (2000); Figure 3.2, case 3). The latter is the most common case, defining what are known as "sink populations" where the intrinsic growth rate r is lower than zero and which can only persist as long as recruitment is maintained through the constant immigration of seeds (plants) or young adults (animals) from neighboring sites that harbor suitable environments and have positive growth rates (source populations; Pulliam, 2000). Finally, the species may also be observed in environments that are unsuitable due to both unsuitable abiotic conditions and negative biotic interactions (Figure 3.2, case 4). This latter case however, represents the least likely unsuitable condition under which a species can sometimes be observed.

A species may also be absent on a site for reasons other than the three main factors discussed so far, such as natural or human disturbances and/or intrinsic population stochasticity, leading to fluctuations of population size in time and space, with possible local and temporary extinctions in some places (Pulliam, 2000). Yet, once extinct locally, whether the species will be able to colonize this same location again once again depends on dispersal limitation (neutral process; Hubbell, 2001). Such fluctuations

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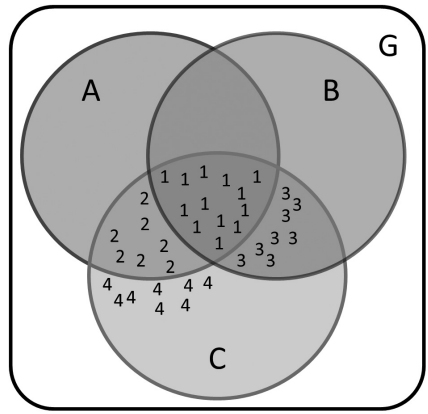


Figure 3.2 The three main factors that drive species ranges. G: studied geographic area; A: suitable abiotic environment (niche); B: suitable biotic environment; C: colonizable range. Observations in the field could result from four situations, from most to least likely: 1. Realized niche (suitable with regard to all three aspects). 2. Suitable abiotic environment with unsuitable biotic conditions, for instance due to strong competition. 3. Colonization outside the suitable environment, maybe due to facilitation (sink). 4. Sink in unsuitable biotic and abiotic conditions, maybe due to historical effect (e.g. trees persisting in unsuitable conditions). Modified from Soberón (2007), with permission.

caused by population dynamics that result in environmentally suitable sites being unoccupied or unsuitable sites being occupied complicate the quantification of species–environment relationships by increasing the percentage of unexplained variation. The question – still largely unresolved for most species – is how much of the variation (if any) in species distributions remains purely stochastic (unexplained variation), once dispersal (for species still undergoing expansion), habitat, and biotic influences have been accounted for? This is even more difficult to assess since population dynamics is itself modulated by these three influences. Dispersal effects (e.g. on the recruitment), the abiotic environment and biotic factors (e.g. the degree of shading for plants or the presence–absence of a mutualist species) can modify the performance of individuals in a population, as well as their birth and death rates.

Despite stochastic fluctuations, species are generally expected to be ruled by a significant local environmental determinism. The quantification of their environmental niches can therefore be expected to provide information for predicting their distribution in environmental, and thus also in geographic, space. The rather unlikely exception to this would be for species where the fluctuations in population dynamics and the strength of biotic interactions would be so high that these processes

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would override, and thus entirely hide, the species–environment relationship. On very large spatial scales, the distributions of nearly all species seem to be determined by the abiotic environment, and in particular the climatic conditions. Few species occurring in warm tropical habitats will be observed in cold temperate habitats and vice versa. This means that all species – even the most cosmopolitan and ubiquitous ones, which are found across large distributions in all possible types of habitats – usually exhibit quantifiable climatic preferences (e.g. Ba et al., 2010). The key questions that need to be addressed when defining the link between species distribution and habitat conditions mainly depend on the extent to which proximal environmental gradients (those having causal effects on species) and their variation are included in the analysis. Another key aspect of species–environment responses is that these can also change depending on the spatial resolution used. For instance, responses could change from being narrow unimodal and skewed at 90 m resolution to become piecewise linear and unbounded at 4 km resolution along a gradient of "precipitation in the wettest quarter" for a plant species in California (Franklin et al., 2013). Therefore, the spatial scale of observation and environmental descriptors can also influence our perception and explanation of ecological phenomena (Wiens, 1989; Willis and Whittaker, 2002). These scale considerations will be discussed more thoroughly in Part II, especially in Chapter 8, as the resolution of analysis is important to consider in any modeling exercise and is tightly related to the data used and their availability on various scales. In the following, we will assume that the right resolution and extent has been chosen for the modeled species.

In the following, we will define the three main drivers of species distributions individually – dispersal (3.2), habitat (3.3), and biotic interactions (3.4) – before focusing on the methodological aspects of fitting and evaluating habitat models for the rest of the book (Chapter 4 onwards). For aspects of population dynamics and their variations in space, we refer interested readers to the relevant literature (e.g. Jeltsch et al. (2008) for plants, Buckley (2008) and Morales et al. (2010) for animals; Nenzén et al. (2012) for a spatial population dynamics tool).

3.2 Speciation, Dispersal, Species Pools, and Neutral Theory

All species have emerged from evolutionary processes at some time and place on Earth. How and where are questions that can help explain patterns of biodiversity, in space on a local, regional, or global scale, but

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also over time (through e.g. studying bursts of diversification; Willis and McElwain, 2002). There are numerous causes of speciation, the two most commonly discussed being allopatric and sympatric speciation (Doebeli and Dieckmann, 2003; Wiens and Graham, 2005). Allopatric speciation arises when geographic barriers emerge that split the initial range of an ancestor species into two or more disconnected sub-ranges, leading to a disruption in gene flow between the separated populations, and ultimately to speciation into two or more distinct taxa (species or subspecies) on each side of the barrier (Hoskin et al., 2005; Grant and Grant, 2009). Sympatric speciation arises when divergence between populations take place within the initial range of the ancestor species, usually due to the ecological specialization of some populations to distinct environmental conditions (e.g. Filchak et al., 2000; Tautz, 2003). Intermediate or different speciation models also exist (parapatric speciation, quantum speciation, etc.).

For the purposes of introducing habitat suitability modeling, it is only necessary to know that such geographic or ecological speciation processes have occurred and are continuing to happen, though at a very slow pace. More important for habitat suitability modeling are the consequences such allopatric and sympatric speciation types may have in terms of shaping past, present, and future patterns of species distributions. For instance, do species that have resulted from sympatric speciation tend to be ecologically or behaviorally more specialized than species resulting from allopatric speciation? To what extent do species retain the ecological characteristics of their ancestor species after divergence (i.e. phylogenetic niche conservatism; Münkemüller et al., 2015) and what role does niche conservatism play in speciation processes (Wiens and Graham, 2005)? These are important issues to address in order to understand current distributions (Pearman et al., 2008a), with potentially important applied implications, such as the effect niche changes between native and invaded ranges can have on our capacity to predict and anticipate invasions (Guisan et al., 2014).

What we learn from evolutionary processes is that many species are today found in a given area (such as e.g. a subcontinent) primarily because they originated there or nearby (Cox, 2001). On such very large scale, biogeographic history and dispersal limitation predominate, with environmental suitability playing only a secondary role in explaining where on the globe the species is currently found. Worldwide biological invasions provide clear evidence of dispersal limitation for most floras and faunas. When brought to a new area, a certain proportion of the introduced

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species will establish and reproduce even in a new environment (alien or exotic species). In plants, we distinguish the biological invasions that began with the first expeditions across the oceans (around 1500; now referred to as neophytes) from the invasions that occurred even earlier (referred to as archeophytes). A small proportion of the established species spread successfully across the landscape (invasive species; Richardson et al., 2000), ultimately affecting native communities and ecosystems and causing damage to the human economy (e.g. in agriculture) or health (e.g. allergenic plants or pathogens). We learn from these processes that maladaptation to the environment is only one of several possible causes for the absence of a species in one specific location, dispersal limitation being another alternative.

The pool of species that are present in, or have dispersed to, a region is thus an important foundation for the types of communities and ecosystems that can be assembled from them (Zobel, 1997; Aarssen and Schamp, 2002; Graves and Rahbek, 2005), illustrating that local species diversity depends, at least partly, on regional species diversity (Caley and Schluter, 1997). Large-scale evolutionary processes that shape regional species pools therefore also have an impact on the expected local diversity, defining the source pool of species for a given site. The latter thus results from the balance between speciation, colonization, and local extinctions, and can be used to define the pool of species that can be modeled within a given area (Ricklefs, 1987, 2008; Guisan and Rahbek, 2011).

This idea that dispersal plays an important role in shaping the distribution of species and thus communities is not new. Early biogeographers had already noticed similarities between flora or fauna on separate continents. While some proposed strict long-distance dispersal to explain these patterns, others proposed "past continental bridges" to explain the same pattern, and this at a time when nothing was yet known about plate tectonics and continental drift (see Lomolino et al., 2010). There remains the question as to the extent to which dispersal accounts for observed patterns of species distributions (Svenning and Skov, 2007)? Taking an extreme view, one might wonder as to whether dispersal alone could account for all the observed variation in species composition, thus asking: "are communities mostly shaped by dispersal?" This is part of the view (together with speciation and extinction) taken to define the unified neutral theory of biodiversity and biogeography (Hubbell, 2001), which can be used as a null hypothesis to test the effect of the environment in shaping species distributions (i.e. neutral patterns assume neutral, equivalent environmental niches across species).

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These dispersal-related aspects can be used as the basis to define the accessible area for a given species (Barve et al., 2011), at a given time, as the whole area within the colonizable reach of existing populations, i.e. where migration is not impeded by natural or human-made barriers to dispersal. For the future conditions, it encompasses those areas that can be naturally colonized in the future, e.g. if species change their distribution in response to climate change or following invasions. Historic factors (e.g. glaciations, climate shifts, etc.) are thus accounted for in the accessible area concept. The latter can reveal possible dispersal limitations and limited range filling, when the observed distribution range is smaller than the full accessible area (e.g. Svenning and Skov, 2004).

3.3 The Abiotic Environment: Habitats and Fundamental Niches

How does the abiotic environment influence the distribution of organisms? What are the different types of environmental influences on species distributions? How do multiple variables jointly determine a species' geographic distribution? These are the main questions that underpin this book, as they all relate to assessing species' "habitat suitability."

Early biogeographers (e.g. Von Humbolt or Darwin) observed that a same species could occur in sites with different environmental conditions, each single combination representing a distinct habitat (sensu Kearney, 2006; see Glossary), and thereby occupy a range of different habitats. Yet, most species have limited geographic ranges (Woodward and Kelly, 2003). From this, one can deduce that a species can colonize a range of conditions along environmental gradients, but that in most cases this range only occupies a proportion of all the possible habitat conditions available, resulting in the species occupying only a limited geographic and environmental range where these specific conditions are met. Why is this so? Why aren't all species found everywhere?

The answer lies in the physiology of organisms (Woodward and Kelly, 2003) and the specialized physiological adaptations that most species have undergone through evolution in order to survive and be competitive in specific habitats. These adaptations come at the cost of being maladapted (and thus unable to survive) or less competitive (and thus excluded) in other habitats, representing adaptation or functional tradeoffs (Woodward, 1987; Kearney and Porter, 2009). The physiological specialization usually results in different shapes of responses being expected for different species along environmental gradients (Whittaker, 1967;

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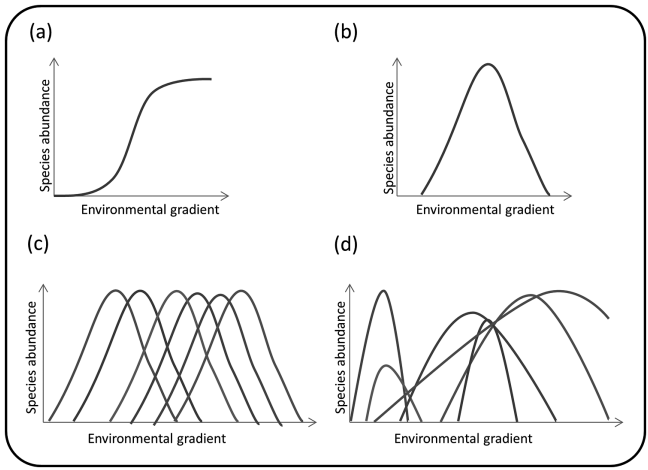


Figure 3.3 Fundamental response curves of species along a hypothetical environmental gradient. (a) Typical sigmoidal species response curve, usually expected at the end of gradients or resulting from the truncation of a unimodal curve. (b) Unimodal response curve. (c) Species packing hypothesis (unimodal responses for several species with regular placement of optima). (d) The response curves of different species vary in shape, amplitude and width (inspired from Austin and Gaywood 1994; Austin 2002, with permission). See Figure 3.4 and Section 3.4 for how fundamental curves can be modified by biotic interactions.

Austin, 1985; Austin and Gaywood, 1994). In most cases this is associated with a position along the gradient where the species performs best – the physiological optimum – and a gradual decrease in performance the further one moves away from this optimum, in either direction (Ellenberg, 1953, 1954; Hector et al., 2012). Such physiological response curves can therefore be represented as sigmoidal or unimodal shapes (Figure 3.3), with unimodal responses appearing to be dominant in nature (e.g. linear responses can be observed for species at the end of gradients or along very stressful gradients). The width of the curve also documents the physiological tolerance of species along the gradient. Different species have different optima and tolerance along a same gradient (Figure 3.3c), from narrow to wide. A species can have a wide tolerance along one variable but a narrow tolerance along another. A species that has a very broad tolerance along a specific gradient may be considered indifferent to variations in this variable, and thus be considered a generalist species (for that gradient).

The transition from optimal to poor performance can be smooth or abrupt, depending on the types of physiological mechanisms involved. An

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abrupt transition can result from limiting factors with threshold effects below or above which some metabolic pathways (e.g. photosynthesis) abruptly change. For instance, in conifer trees, the cambium activity allowing root growth tends to stop rather abruptly below some threshold value of soil temperature (around 7°C at -10 cm for Arolla pine *Pinus cembra* in the Alps; Körner, 2003; Körner and Paulsen, 2004). Smoother transitions occur when the gradient has a more or less linear effect on some metabolic rates (e.g. carbon sequestration, water use efficiency), progressively lowering individuals' fitness.

When looking at the effect of each environmental variable independently, one may fail to identify the combined effects (i.e. interactions) of some of these variables on the species' physiology, one possibly dampening or amplifying the effect of another. Therefore, all important variables should be considered jointly in an analysis, in order to define what is known as the environmental niche of species (Chase and Leibold, 2003), a term initially coined by Grinnell (1917). When considered jointly, the physiological responses of a given species to several environmental variables define a multidimensional volume called a species' fundamental environmental niche (see Figure 3.5a). The fundamental niche concept was quantitatively formalized by Hutchinson in (1957), based on earlier insights from Grinnell (1917) and others (see Chase and Leibold, 2003 for a review of the niche concept in ecology). Hutchinson defined it as an n-dimensional hypervolume (i.e. of possibly $n > 3$ dimensions) in a space defined using environmental variables hypothesized to have a direct influence on a species' physiology, and within which the population growth rate is positive (see Pulliam, 2000; Kearney, 2006). Hutchinson illustrated the concept with animal species, but the same concept applies to plants and other organisms (i.e. fungi, protista, bacteria), and Ellenberg had already proven this concept experimentally with graminoid plants along a water table gradient a few years before Hutchinson's conceptual summary (Ellenberg, 1953, 1954; Austin, 1990; Hector et al., 2012). For plants, the axes of the fundamental niche may typically be resource variables related to light, heat, water and nutrient availability, or regulators such as too high or too low temperatures or other extreme climatic conditions (Guisan and Zimmermann, 2000). For animals, they may be thermal limits, and/or water/food/habitat availability (Kearney and Porter, 2009).

If one were able to identify all the important environmental variables for a species, quantify its fundamental niche (i.e. physiological responses) along these gradients, and have these environmental variables available as global maps, one could predict all the geographic

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locations where the species could establish and maintain viable populations in the hypothetical absence of biotic interactions (Clark et al., 2007), and provided the species can access all locations (Barve et al., 2011). In practice, however, measuring the fundamental niche based on field observations in natural conditions is practically impossible, especially since most species require interactions with other species (e.g. pollinators, symbionts, etc.) to persist. As a result, it is often stated that only "mechanistic" models based on measured physiological or behavioral parameters (e.g. Kearney and Porter, 2009), or analyses based on *ex situ* data (e.g. a plant grown in botanical gardens outside its natural range; Vetaas, 2002), can approach the fundamental niche. There are two main reasons for our inability to capture the fundamental niche from field observations. First, the responses of species in nature are modified by their interactions with other species, within the same group (e.g. competition for light in plants or for food resources in animals) or across groups at different hierarchy or trophic levels (e.g. predation, parasitism, etc.), combined with the effects of migration limitation in response to changing environments (land use, stochastic disturbances). The biotic component, with the different types of possible interactions, will be discussed extensively in Section 3.4. These effects constrain the fundamental niche to what is considered to be the realized niche, the only one that can be observed from field observations (see Section 3.4 for an extended coverage of this aspect). Second, it is not always possible to spatially map and measure the variables that have a significant effect on a species' physiology (e.g. minimum absolute temperature during the relevant part of the year for a plant or animal, exact soil moisture for a plant). This is often due to material limitations (i.e. no system capable of measuring the target variable in the wild, or too costly to monitor numerous individuals in numerous populations), or due to our inability to map such important variables precisely, and to link species observations to these physiologically important predictors.

Furthermore, knowing which variables have a direct physiological effect (i.e. should be used to define the fundamental niche) requires, for each species, prior experimental laboratory measurements (e.g. measuring metabolic rates and individual fitness while varying environmental variables), which is not feasible for all species. This becomes even more difficult if such responses are measured over a large number of populations to account for genetic differences among populations. It is therefore often easier to use measurements of surrogate environmental

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variables that are hypothesized to best correlate with the physiologically meaningful variables (e.g. altitude for temperature, or minimum of mean monthly temperature for absolute minimum temperature), but with the consequence to reduce both the predictive power (e.g. for projections in space or time) and the level of generalization of the model (e.g. altitude cannot be compared between population in cold environments in the Alps and the Arctic; Pellissier et al., 2013a). These surrogates can be ranked along a scale of "proximality," depending on how close they are to having a causal effect which explains the distribution of the species studied (Austin, 2002, 2007). The different variables can also be classified, depending on their effect or use by the target species, as limiting factors (causing linear or step responses), regulators (modulating the organism physiology, gradual response) or resources (consumed by the organism) (Austin, 2002; Huston, 2002; Austin, 2007; and see Section 4.1).

3.4 The Biotic Environment: Species Interactions, Community Assembly, and Realized Niches

The biotic environment covers all possible forms of interactions among species, either within the same or between different trophic levels. It can also further constrain a species' distribution in space, in time, or along environmental gradients, in addition to the abiotic constraints. For the purpose of simplification, in these examples we often start by considering competition because historically it has been the first biotic factor discussed by the proponents of the environmental niche concept (e.g. Hutchinson). However, interactions between trophic levels (e.g. predator–prey, plant–pollinator, or plant–herbivore) or functional groups (e.g. host–parasite, symbiosis) can be as important in many instances (e.g. Broitman et al., 2009; Pellissier et al., 2013d; Lira-Noriega and Peterson, 2014). Take the fundamental niche of species as described in Section 2.3. It considers that a species can occupy all locations where the abiotic, environmental conditions allow the species to maintain viable populations. However, we also saw earlier on that species do not usually have uniform responses along environmental gradients, and thus the fitness of their individuals, and potentially their competitive potential, also varies along these gradients. One direct implication of this pattern is that when two species co-occur at a same location and compete for the same resource ultimately the less fit species will be excluded, as predicted by the "competitive exclusion principle" (Salisbury, 1929; Gause, 1936). There is extensive evidence

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of this general principle (e.g. Ellenberg, 1953; Brown, 1971; Grace and Wetzel, 1981; see Lomolino et al., 2010). One piece of evidence, to be further investigated, may be the fact that a number of high-elevation alpine plants can be grown at low-elevation botanical gardens (i.e. in warmer parts of the temperature gradient) in the absence of competition. However, they are never observed at these low elevations in natural systems, probably because they are systematically outcompeted by higher stature plants optimized for low elevations (e.g. Vetaas, 2002). Of course, this example remains speculative for mountain plants, which require some of their environmental conditions to be physically engineered (e.g. watering by gardeners) in order to survive at low elevations, as these may be the actual factors defining their lower limit. The signatures of such competitive exclusion was generally considered to be only observable at local scales (e.g. Pearson and Dawson, 2003), where individuals compete, but evidence has recently shown that the signatures of competitive exclusion can also be detected on larger scales (Gotelli et al., 2010).

When a species is systematically excluded from parts of an environmental gradient (as is putatively the case for alpine plants), the response along these gradients revealed from field observations only gives a partial view of the full physiological (i.e. fundamental) response. The species response from field observations along gradients thus depends on the particular biotic configuration in the area studied. This type of field observation-based response along a single gradient has been called the "realized response" or "ecological response" (Austin et al., 1990; Austin and Gaywood, 1994), in contrast to the "physiological response" or "fundamental response" described above. The differences between these two types of response have also been demonstrated experimentally, as for instance by Ellenberg (1953) with monocultures and mixtures of plants along a gradient of increasing depth of water table. Depending on which part of a fundamental response is excluded by competition, different shapes are obtained for the realized response. Walter (1960; in Ellenberg, 1988), Austin (2002) or Hector et al. (2012) illustrate different cases, such as exclusion of the optimum leading to a bimodal realized response, exclusion of a margin leading to a truncation of the realized response and displacement of the optimum, or exclusion at both margins leading to a similar optimum, but a narrower realized response (Figure 3.4).

Interestingly, it was observed very early on that the fundamental niche of species is more constrained by competition in the milder parts

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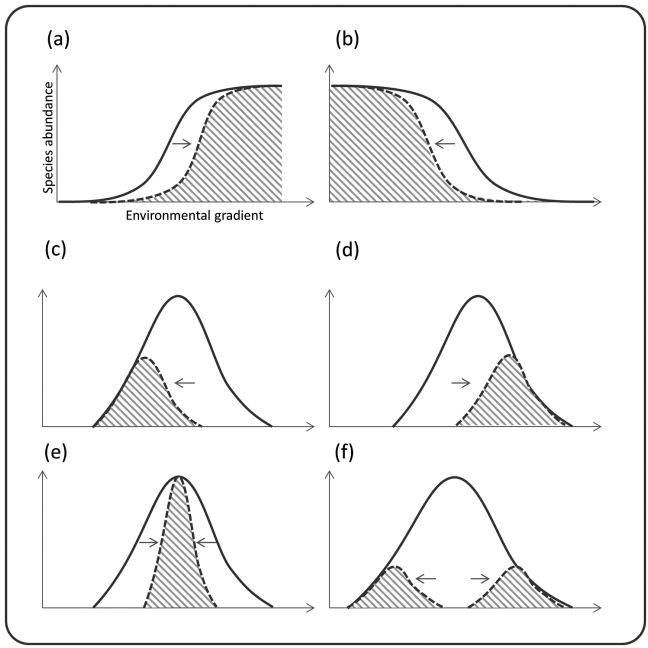


Figure 3.4 The main situation of competitive exclusion constraining the fundamental response curve of a hypothetical species along a hypothetical environmental gradient into the realized response curve observed in natural systems (i.e. niche truncation). These modifications include linear response displacement (a, b), optimum displacement (c, d), narrowing of the niche (e), and optimum removal (resulting in a bimodal response) (f). Modified with permission from Austin and Smith (1989) and Austin (1990).

of environmental gradients that require less physiological adaptation and thus provide better growth conditions (e.g. at the warm end of a niche) than at the gradient edges with higher physical stress (Austin and Gaywood 1994). This resulted in the proposal of a general rule stating that the ranges of species along environmental gradients tend to be limited by physiological tolerance toward the physiologically more stressful edge, and by competitive interactions (i.e. exclusion) toward the physiologically less constraining and more productive parts of environmental gradients (Connell, 1961; Austin et al., 1990; Brown et al., 1996), a hypothesis recently named the "asymmetric abiotic stress limitation" (AASL) hypothesis (see Normand et al., 2009; Meier et al., 2011). This has, for instance, been shown experimentally for cattails along a water depth gradient (Emery et al., 2001) and for chipmunks along an elevation gradient (Heller and Gates, 1971); see Smith and Smith (2015).

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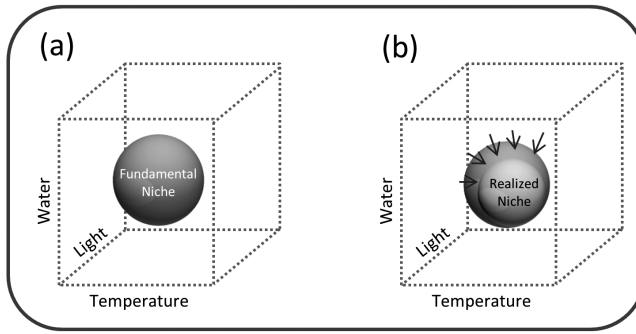


Figure 3.5 Illustration of a three-dimensional environmental niche for a hypothetical species. (a) Fundamental niche representing the envelope of environmental conditions (envelop) within which a species can maintain a viable population in the absence of competitive interactions. (b) Realized niche, representing a subset of the fundamental niche constrained by negative interactions (e.g. competition) with one or several other species. With more than three dimensions, the niche can no longer be represented graphically and is called a hypervolume.

The distinction between the two types of niche responses along single gradients leads to the same distinction at the level of the whole environmental niche, which was termed the "realized niche" by Hutchinson (1957) or the "ecological potential" by Ellenberg (1953), describing a subset of the fundamental niche (or physiological potential) constrained by competition with one or several other species (Figure 3.5).

We have now seen that biotic interactions constrain the fundamental niche thus forming the realized niche. Numerous different biotic interactions can affect the predictability of a species at a site from environmental predictors only (Araújo and Guisan, 2006; Sutherst et al., 2007; Elith and Leathwick, 2009; Kissling et al., 2012; Wisz et al., 2013). These biotic interactions may either be negative, by excluding a species from sites that are a priori environmentally suitable (i.e. within its fundamental niche; see above; e.g. competition) or facilitate a species at sites that appear environmentally unsuitable based on measured average site conditions (i.e. depending on the scale of measurement; Pellissier et al. (2010)).

Examples of positive interactions (i.e. facilitation; Boucher et al., 1982; Callaway, 1995; Stachowicz, 2001; Bruno et al., 2003) include commensalism, mutualism (e.g. non-symbiotic, Pellissier et al., 2012a; or symbiotic, Pellissier et al., 2013c), biotic engineering (i.e. a species improving the micro-habitat conditions for another), for instance forest understory species that cannot grow in plain light benefiting from the shade from the

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surrounding tree canopy (Linder et al., 2012; Nieto-Lugilde et al., 2015), some plant species benefiting from fungi (i.e. mycorrhizae) or bacteria (i.e. nodules) that capture atmospheric phosphorus or nitrogen in nutrient-depleted soils (e.g. Defossez et al., 2011; Pellissier et al., 2013c), or parasites requiring the presence of their host (e.g. Olwoch et al., 2003). Examples of negative interactions are more common and include competitive exclusion (e.g. Leathwick and Austin, 2001; Hakkarainen et al., 2004; Meier et al., 2010; Pellissier et al., 2010; Meier et al., 2011; Meier et al., 2012), predation (incl. herbivory; e.g. Pellissier et al., 2012b), or parasitism when the host is sufficiently affected by its parasite to be removed from entire parts of its range (e.g. Olwoch et al., 2003). Many of the previous examples of interactions often involved pairs of species (or functional groups of species), but interactions also naturally take place within more complex biotic frameworks such as food webs and large interaction networks (Schoener, 1989; Polis et al., 1997; Bascompte et al., 2003; Sargent and Ackerly, 2008; Ings et al., 2009). From a geographic perspective, the interactions derived from these interactive systems can be used as predictors of individual species distributions (Gravel et al., 2011; Pellissier et al., 2013d; Albouy et al., 2014). The way biotic interactions influence the presence (positive interactions), absence (negative interactions) or abundance (both) of a given species therefore also influences the final composition of a community (Weiher and Keddy, 2001; Lortie et al., 2004; Ricklefs, 2008; Guisan and Rahbek, 2011; Kissling et al., 2012; Wisz et al., 2013), as will be discussed later (Section 4.3). Quantifying these interactions into assembly rules is therefore required in order to understand how communities assemble, and – if the quantified rules allow – to ultimately predict assemblages (Weiher and Keddy, 2001; Fraser and Keddy, 2005; Ricklefs, 2008; Guisan and Rahbek, 2011; Hortal et al., 2012).

3.5 Further Discussion of the Realized Environmental Niche and Other Related Niche Concepts

We have mainly relied so far on Grinnell's and Hutchinson's concept of the environmental niche. However, several other concepts exist. Leibold (1995), in a review, and later on Chase and Leibold (2003), present two opposing views of the niche: (i) determined by the environmental *requirements* of species, i.e. consistent with Hutchinson's definition of the environmental niche; or (ii) determined by the *functional role* a species plays within a food chain and the *impact* it has on its environment (mainly

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the resources it consumes), i.e. consistent with Elton's (1927) definition of the niche. Hence, the former is embedded within an autecological and physiological view of the niche (*environmental niche*; see e.g. Austin, 1992), whereas the latter opts for the trophic levels and food web theory perspective (*trophic niche* as termed by Elton; see Austin et al., 1990; Silvertown, 2004). The two concepts are both relevant in nature, yet are often considered independently in the contemporary literature.

The habitat models literature addressed in this book, mostly focuses on the environmental niche concept. However, part of the trophic niche is nevertheless implicitly included in the restriction of the fundamental environmental niche into the realized niche and recent developments have linked food webs (and thus trophic relations) with habitat distribution modeling, demonstrating that the two concepts are intertwined (Pellissier et al., 2013d; Albouy et al., 2014). After the early work by Grinell, Hutchinson and Elton, further contributions to the niche concept were made by MacArthur (1968, 1972) and colleagues, resulting in numerous refinements, complications, and clarifications, published in the interim (e.g. Colwell and Futuyma, 1971; Vandermeer, 1972; Whittaker et al., 1973; Austin et al., 1990; Leibold, 1995; Pulliam, 2000; Chase and Leibold, 2003; Kearney and Porter, 2004; Silvertown, 2004; Wiens and Graham, 2005; Araújo and Guisan, 2006; Kearney, 2006; Leibold and McPeek, 2006; Soberón, 2007; Pearman et al., 2008a).

One proposed refinement – still not generally accepted – was to differentiate between α and β niches (and associated α and β traits) according to the scales at which measurements are made and at which niche occupancy operates. According to this distinction, the α-niche would represent the realized niche of a species, which would correspond to the niche built from environmental measurements made at the local community scale at which interactions among species occur (Pickett and Bazzaz, 1978; Silvertown, 2004). It would thus represent field observations and abiotic physico-chemical measurements at the level of conditions occupied by individuals of a species with respect to other individuals of the coexisting species in a community, assuming that differences between species measured at this scale should facilitate the species' coexistence (Silvertown, 2004; Silvertown et al., 2006). In contrast, the β-niche would correspond to the niche built from measurements at the level of mean habitat conditions within a community (e.g. mean annual temperature of a vegetation plot), and was initially defined as measures making it possible to differentiate environmental conditions between communities along gradients (Silvertown et al., 2006).

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Although clearly of interest and worth further developments, these two niche measurement levels have been so far developed by only a few authors, and therefore should be treated with caution before applying them too widely. When considering these concepts in an HSM context, most analyses and models built by relating species occurrences to attributes of environmental variables therefore represent calibrations of the β-niche, while the α-niche can only be approached through local studies with exhaustive sampling of whole communities within plots and measurements of the physical environment at a sub-pixel or micro-scale (M. D'Amen and A. Guisan, unpublished). It can therefore be assumed that we will mostly address issues related to the β-niche in this book, but we will refer to the "niche" throughout for the purposes of simplification. With the latest advances in very high-resolution data coupled with intense field sampling (e.g. 1 or 2 m; Lassueur et al., 2006; Pradervand et al., 2014), however, we may get closer to the α-niche in the future (D'Amen and Guisan, unpublished).

Another recently-defined concept concerns the "potential niche". This represents the part of the fundamental niche that is actually available to species as constrained by the realized environment (Jackson and Overpeck, 2000; Ackerly, 2003). Not all the possible combinations of a set of environmental variables that make up the fundamental niche actual exist in a study area (e.g. Austin et al., 1990), or possibly even on Earth (see also Section 6.3 for some examples). In this context, another useful term is that of the "realized environment," which describes the combination of all the environmental conditions that do exist in the study area (see Guisan et al., 2014). The potential niche thus slightly differs from Hutchinson's niche in that it is constrained by what is actually available to the species in the area studied. Thus, in practice, when fitting the realized niche from field observations, one implicitly takes into account the constraints of the potential niche, provided that the sampling is representative of the whole study area. The realized niche can thus be given a more complete working definition as the fundamental niche constrained by biotic interactions, dispersal, but also the realized (available) environment. This is the definition we will use throughout the book when referring to the realized niche. It should be noted that we find the term "potential niche" rather misleading in the context of HSMs, since we refer to the model-based, projected distribution of a species as the "potential distribution", thus accepting that some parts of this mapped distribution are actually not colonized by the species due to dispersal constraints, random effects, local peculiarities not captured in

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the predictors, etc. However, the potential distribution of a species refers to the realized niche, not to the fundamental or potential niche.

Finally, we should remember at this stage that the original definition of the niche implies a fitness component, i.e. within the niche the species should be able to maintain positive population growth rate. In practice, observations in the field may include sink populations under conditions outside of the fundamental niche of a species (i.e. unsuitable habitats) but which are maintained through high propagule pressure (i.e. dispersal) from nearby source populations under conditions inside the fundamental niche (Pulliam, 2000). A direct corollary of this is that quantifying the realized niche from empirical field observations requires conducting a thorough sampling of the local populations and measurements need to include fitness parameters so that inappropriate sink populations can be identified and excluded from the dataset before quantifying the niche (Pellissier et al., 2013b). If this is not possible (e.g. if only occurrences or presence–absence data are available), then only an approximation of the realized niche is obtained, which may suffice to produce appropriate predictions in many cases (e.g. raw predictions of climate change effects) but not in others (e.g. conservation actions for sensitive species). In these cases, habitat suitability maps derived from presence–absence data may not provide information on population demography but rather on the species' frequency in those habitats (Thuiller et al., 2014b). Most of the predictions of species distribution found in the literature are indeed based on such approximations of the realized niche. Accidentally including a few sink populations will not drastically change the estimation of the realized niche if a fairly high number of samples from fit populations have been collected.

To sum up, the realized niche fitted in HSMs can be viewed as the intersection of conditions suitable to the species in terms of: (i) the abiotic conditions within the available environment (i.e. the potential niche), (ii) the biotic conditions, and (iii) the accessible conditions (i.e. within the colonizable or accessible range; see Section 4.2) (Pulliam, 2000; Soberón, 2007; Soberón and Nakamura, 2009). Furthermore, HSMs often fit this realized niche by relating presence-only or presence–absence observations to average environmental conditions within the modeled cells, and thus tend to ignore population fitness and to fit the niche at the β-level. Future HSM research should tend to use population fitness data more systematically, and therefore sampling such data in the field should be encouraged.

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